1. SJF — Shortest Job First

```
n = int(input("Enter number of processes: "))

processes = []

for i in range(n):
    bt = int(input(f"Enter Burst Time for P{i+1}: "))
    processes.append([i+1, bt])

processes.sort(key=lambda x: x[1])

time = 0
total_wt = 0
total_tat = 0

print("\nProcess\tBT\tWT\tTAT")

for p, bt in processes:
    wt = time
    tat = wt + bt

    print(f"P{p}\t{bt}\t{wt}\t{tat}")

    total_wt += wt
    total_tat += tat
    time += bt

print("\nAverage Waiting Time =", total_wt / n)
print("Average Turnaround Time =", total_tat / n)
```

2. RR — Round Robin

```
n = int(input("Enter number of processes: "))

bt = []

for i in range(n):
    bt.append(int(input(f"Enter Burst Time for P{i+1}: ")))

q = int(input("Enter Time Quantum: "))

remaining = bt.copy()
wt = [0] * n
time = 0

while True:
```

```
    done = True

    for i in range(n):
        if remaining[i] > 0:
            done = False

            if remaining[i] > q:
                time += q
                remaining[i] -= q
            else:
                time += remaining[i]
                wt[i] = time - bt[i]
                remaining[i] = 0

    if done:
        break

tat = [wt[i] + bt[i] for i in range(n)]

print("\nProcess\tBT\tWT\tTAT")

for i in range(n):
    print(f"P{i+1}\t{bt[i]}\t{wt[i]}\t{tat[i]}")

print("\nAverage Waiting Time =", sum(wt) / n)
print("Average Turnaround Time =", sum(tat) / n)
```

3. LRU — Least Recently Used

```
pages = list(map(int, input("Enter page reference string: ").split()))
capacity = int(input("Enter number of frames: "))

frames = []
page_faults = 0

for page in pages:

    if page not in frames:

        page_faults += 1

        if len(frames) < capacity:
            frames.append(page)
        else:
            frames.pop(0)
            frames.append(page)
```

```
    else:
        frames.remove(page)
        frames.append(page)

    print("Page:", page, "Frames:", frames)

print("\nTotal Page Faults =", page_faults)
```

4. MRU — Most Recently Used

```
pages = list(map(int, input("Enter page reference string: ").split()))
capacity = int(input("Enter number of frames: "))

frames = []
page_faults = 0

for page in pages:

    if page not in frames:

        page_faults += 1

        if len(frames) < capacity:
            frames.append(page)
        else:
            frames.pop(-1)
            frames.append(page)

    else:
        frames.remove(page)
        frames.append(page)

    print("Page:", page, "Frames:", frames)

print("\nTotal Page Faults =", page_faults)
```